PDANet: Point Distribution Awareness for 3-D Object Detection From LiDAR Point Clouds

Miao Tang, Dianyu Yu, Qingyong Hu, Wenxia Dai, and Wen Xiao, *Member, IEEE*

***Abstract*— Recent deep learning models for 3-D object detection (3DOD) based on LiDAR point clouds have exhibited promising performance in the detection of on-road objects. However, their performance is often constrained when detecting small objects like pedestrians and cyclists. Due to occlusions and their inherent volumes, the spatial and semantic information of small objects is relatively limited. Therefore, we propose a point distribution awareness network (PDANet) that emphasizes the structural distribution of objects, to enhance the extraction of positional, geometric, and semantic features. It comprises two modules: a multifeature adaptive perception (MFAP) module, enhancing local feature perception, and a semantic-aware downsampling module, enhancing the selection of foreground points. These modules can be used in a plug-and-play manner for point downsampling feature perception. They are then integrated as a branch into representative point-based (IA-SSD), point-voxel-based (PV-RCNN), and voxel-based (Centerpoint) detectors, resulting in PDA-SSD, PDA-PV-RCNN, and PDA-CenterPoint models. Through experiments on the ONCE dataset, the PDANet brought up the mean average precision (mAP) of all three types of baseline models by 3.03%, 4.60%, and 6.15%, respectively. Further experiments on the KITTI dataset illustrate an improvement in average precision (AP) of small objects up to 7.41%.**

***Index Terms*— 3-D object detection (3DOD), plug-and-play module, point clouds, small object.**

I. Introduction

MULTIOBJECT detection based on LiDAR point cloud has been increasingly studied for the perception of 3-D environments, for applications such as autonomous vehicles and intelligent infrastructure [1]. The objective of 3-D object detection (3DOD) is to determine the position, size, and category of key objects, such as cars, pedestrians, and cyclists in road environments. With the recent advancement of deep learning techniques in computer vision, 3DOD from LiDAR point clouds has been rapidly developed [2], [3], [4] and achieved increasingly better performance on public datasets such as KITTI [5] and ONCE [6]. Efficiently representing sparse and unordered point clouds is crucial for 3DOD. The sparsity of point clouds is a key characteristic that sets them apart from 2-D detection [7], [8], and they cannot be directly processed with the now mature 2-D network models [9]. Some methods utilize PointNet [10] and PointNet++ [11] to directly learn 3-D representations [1], [12], preserving the geometric features of the original point cloud as much as possible. Such methods, which operate directly on raw point clouds, are referred to as point-based methods that accurately estimate object positions and shapes. Similar to 3-D segmentation [13], [14], current point-based methods [15], [16], [17] have shown that capturing more detailed spatial geometry and semantic structures can improve the accuracy of 3DOD. Pointformer [18] employs the Transformer module to learn contextual features of the scene. GAPSA [19] learns beneficial features in region of interest (RoI) using a gateway attention mechanism and RoI pooling module. Nevertheless, existing literature primarily focuses on input point coordinates and their corresponding feature in the feature aggregation stage, without exploring effective local multifeature information [12], [20], [21].

The feature extraction module undergoes multilayer downsampling, aiming to reduce the computational cost of a large number of point clouds while extracting as many object features as possible. However, the downsampling process reduces the number of object points in the scene, which can compromise detection performance. To tackle this issue, one strategy is to increase the recall rate of foreground points during the sampling process, which helps to learn as much information as possible from RoI. 3DSSD [22] uses feature distance as a sampling criterion, adaptively increasing the recall rate of foreground points. SASA [12] and IA-SSD [23] use discriminative scores during downsampling to distinguish the importance of each point. SSA3D [21] employs multiple attention sampling strategies to sample high-quality points and improve the detection accuracy of vehicle categories on the KITTI dataset. Current sampling methods [21], [24] pay less attention to small objects, like pedestrians [6]. Solely sampling foreground objects can often lead to the loss of geometric information for such small objects.

Besides the employed point-wise representation in 3DOD, methods based on voxel [25] and point-voxel [24] have also achieved notable detection performance. The voxel-based methods employ a voxel representation of the 3-D space, and the features learned through 3-D convolution are often

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converted into 2-D BEV feature maps for the final object classification and boundary box regression. The point-voxel methods can perform feature fusion through 3-D convolution, BEV mapping, and then RoI pooling. However, voxelization and BEV representations inevitably result in the loss of localization information [19].

To address these limitations, a point distribution awareness network (PDANet) is proposed, which comprises a multifeature adaptive perception (MFAP) module that explores beneficial features in the local neighborhood, and a semantic-aware downsampling module that preserves shapes of small objects, enhancing the model's ability to perceive object shapes and positions. The PDANet can also function as a plug-and-play branch for point feature extraction tasks, supplementing the information loss caused by voxel-based networks.

In short, our contributions of this article are as follows.

1) Introducing a PDANet to achieve critical foreground point sampling and feature enhancement, improving the overall accuracy of 3DOD.
2) Proposing an MFAP module that utilizes self-attention for adaptive fusion of local multiple features, and a semantic-aware downsampling module to capture small but crucial foreground points.
3) Integrating the proposed PDANet as a plug-and-play branch into point-based PDA-SSD, voxel-based PDA-CenterPoint, and point-voxel-based PDA-PV-RCNN to validate the generality and efficacy in enhancing overall detection performance.

II. Related Work

A. 3-D Point-Based Detectors

Point-based detection is now a mainstream method for 3DOD [1], [12], [23]. PointRCNN [1] proposes a completely point-based detection framework that utilizes point-by-point features to generate an initial 3-D proposal, and in the second stage, bounding box refinement is performed using RoI interior point features. VoteNet [26] is an anchor-free point-based detector that replaces region proposal network (RPN) with a voting scheme that uses lightweight. 3DSSD [22] utilizes multiple set abstraction (SA) layers to form a downsampling feature extraction network and removes the expensive upsampling layer. Recent studies [27], [28] have learned to capture more detailed spatial geometry and semantic information. CT3D [29] enhances the proposal feature representation by utilizing a two-stage attention module, which improves the refinement of the initial bounding box. SSA3D [21] proposes a neighbor attention group module to adaptively obtain the feature importance of the neighboring points around the sampling point, thus helping the model to obtain richer spatial structure information. At the same time, a segmentation auxiliary module implicitly helps the model to learn features better, without increasing the computational cost. GAPSA [19] proposes a gateway attention pooling module to learn discriminative local features from the point cloud. FocusSA [17] proposes a lightweight feature extraction module that considers local geometry, achieving improved performance on the moderately difficult KITTI dataset [5].

B. Density-Based Methods

The inconsistent distribution of points in different areas of a scene presents a challenge for object detection, as some areas may have sparse point clouds that provide limited information about objects. Various strategies have been proposed in literature [30] to address this issue, such as using voxel feature densities in the bounding box refinement stage, as in the PDV method [31]. This approach helps to learn centroid point context information and enhances the model's ability to handle variations in point density. 3-D HANet [32] introduces a 3-D heatmap-assisted network module that improves the model's ability to capture object information without increasing computation in the inference phase. On the other hand, DA-Net [33] incorporates an inverse density scoring module in indoor scenes to weigh the features and direct the network's attention toward the sparse keypoints. Instead, an optimization strategy [30] has been proposed to consider the relationship between point cloud density and distance from the perspective of the data itself. It enhances the performance of existing models by implementing a point cloud density-adjusted preprocessing.

C. Point Downsampling

Downsampling is employed as a crucial step in point-based methods, aiming to enhance the computational efficiency of subsequent stages without loss of scene features. Many point-based methods employ classical FPS in the downsampling process. Although FPS achieves good results, it belongs to unbiased sampling, and because the number of background point clouds in the scene is much larger than that of the foreground point clouds, the FPS operation may result in the loss of useful foreground points [34]. This loss can negatively impact the next feature extraction module. Yang et al. [35] introduces a learnable Gumbel sampling module that computes the importance of each point by utilizing the Gumbel softmax strategy. This allows for the selection of keypoints during the downsampling process. LSNet [36] further utilizes several layers of MLP and one-hot coding operation to achieve competitive performance at 512-point samples. 3DSSD [22] proposes feature distance-farthest point sampling (F-FPS) and Euclidean distance-farthest point sampling (D-FPS) fusion sampling strategy to choose more interesting and important points. SASA [12] makes it easier to sample foreground points of interest by sampling weights through a semantic bootstrap module. IA-SSD [23] relies on the semantic information of the points and proposes an instance-aware method to distinguish between the foreground points and the background points, and at the same time, makes the object point cloud easier to sample, thus the effect of the foreground point sampling has been further improved.

III. Method

The proposed PDANet mainly comprises an MFAP module for local keypoint distribution perception and a semantic-aware downsampling module for global foreground point distribution perception. Inspired by the recent point-based model IA-SSD, PDANet has been used as a feature extraction backbone to

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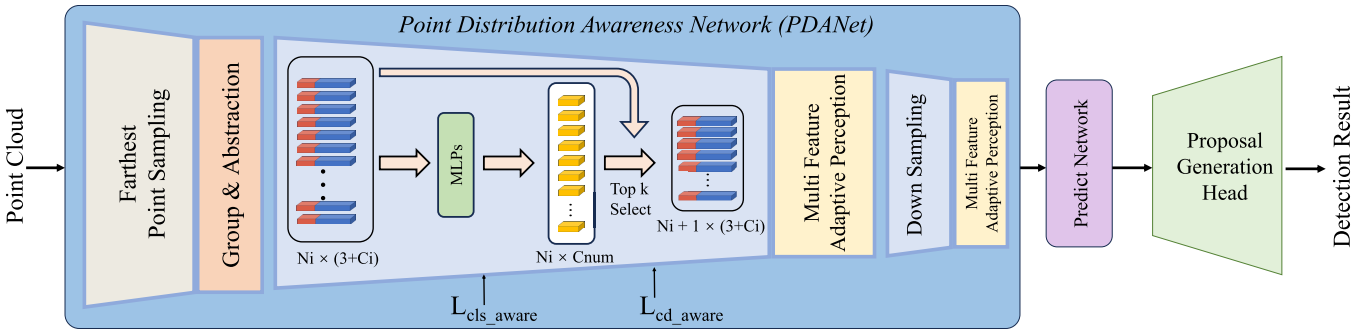


Fig. 1. Architecture of the PDA-SSD model.

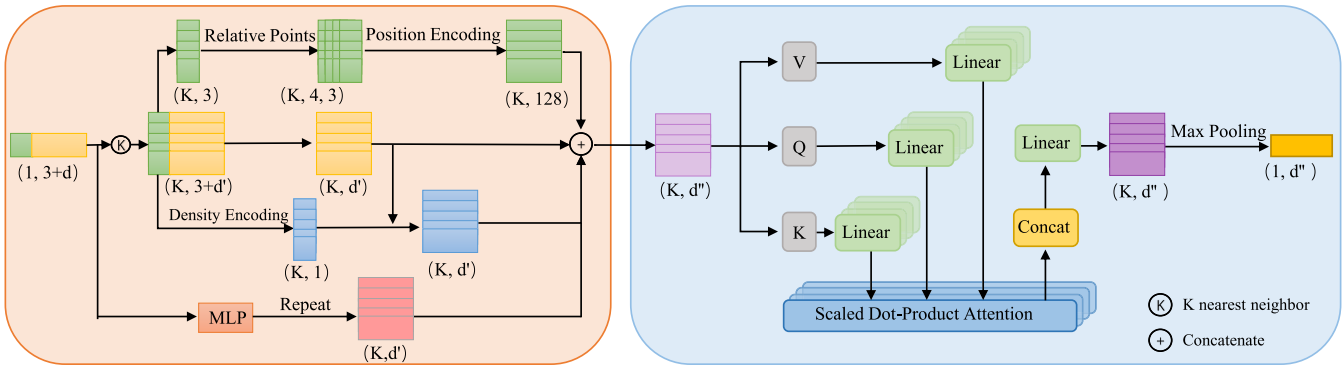


Fig. 2. Overview of the MFAP module. (Left) Multifeature encoding layer, while (right) multifeature aggregation layer.

form a new model, PDA-SSD, as shown in Fig. 1. The PDANet is used to extract keypoint foreground features from the input point cloud. The prediction network utilizes the preserved foreground points to predict the instance center, while the proposal generation head network further obtains the 3-D bounding box and corresponding class labels. To illustrate the plug-and-play nature of the modules, we further detail the integration of the PDANet as an additional branch into the PV-RCNN and CenterPoint models, to complement point-wise feature extraction for improved feature representations.

A. MFAP Module

Existing point-based detection models [12], [21], [23] emphasize input point coordinates and their corresponding features during the feature learning stage. This is achieved through SA operations, as seen in PointNet++ [11], and soft attention mechanisms similar to those in RandLA-Net [37]. Despite their simplicity and effectiveness, these methods lack depth in exploring richer feature information. Regarding point-voxel fusion networks, PV-RCNN [24] employs a PointNet-like [10] architecture within the second stage of RoI refinement for the extraction of both coordinate and semantic features from each keypoint. PDV [31] additionally learns kernel density estimation (KDE) [38], [39] features within the RoI. However, the reliance of this two-stage box refinement mechanism on the initial detection box results in suboptimal performance, particularly evident in small object detection. Hence, an MFAP module is proposed in this article to explore the complementary information surrounding the multiple attributes of keypoints. This module primarily investigates spatial and semantic features within neighborhoods. Spatial features include Gaussian density features that take into account the distribution of point densities and a variety of positional features between different points. Semantic features contain local features of neighboring points and global features of keypoints, where local features facilitate the learning of fine-grained semantic features between points of different domains locally, and keypoint features are represented in a global coordinate system and contain global semantic information about the object. Subsequently, this proposed module adaptively combines their advantages, utilizing an attention mechanism to derive enhanced keypoint features. As shown in Fig. 2, our module is applied to the point downsampling feature aggregation stage, which consists of two modular layers: 1) multifeature encoding layer and 2) multifeature aggregation layer.

1) Multifeature Encoding Layer: The connections between selected points and their neighboring points in a downsampled point cloud involve the acquisition of three distinct features: Gaussian density feature, local spatial feature, and local and global semantic features of the relevant points.

a) Gaussian density feature: Consideration is given to variations in point density distribution within different localized regions. This adaptive approach enables us to accommodate the effects resulting from alterations in localized densities at varying distances and scenes. Inspired by the concept of KDE [31], [38], [39], Gaussian density is employed to assess the distribution of each neighborhood point about the keypoints within a local region. This approach enhances our ability to perceive the different distributions of local neighborhood densities. Gaussian density is defined

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as follows:

$$d_i^k = \frac{\exp\left(\frac{\|p_i - p_i^k\|}{2r^2}\right)}{\sqrt{2\pi}r} \tag{1}$$

where p_i is the keypoint, p_i^k is the keypoint surrounded by the ball query operation in PointNet++ [11], and r is the query radius. It is worth noting that the same point produces different density qualities in different neighborhoods of the keypoints

$$df_i^k = \sigma\left(\mathrm{MLP}\left[d_i^k / \max\left(d_i^1, \ldots, d_i^j, \ldots, d_i^k\right)\right]\right) \odot f_i^k. \tag{2}$$

As shown in (2), to obtain a more precise density score, the density value of neighboring points is calculated. This computed value is then fed into an MLP layer featuring a sigmoid activation function. Subsequently, the corresponding features are re-weighted based on the obtained density score, denoted as df_i^k. This process strengthens the features of low-density points, enabling the module to perceive changes in local point density. Consequently, the module performs better in discriminative features related to small or distant objects.

b) Local spatial feature: For the k neighborhood points of each sampling point $P_i^1, \ldots, P_i^k, \ldots, P_i^K$, the following relative positional coding is executed:

$$r_i^k = \mathrm{MLP}\left(P_i \oplus P_i^k \oplus \left(P_i - P_i^k\right) \oplus \left(\frac{P_i - P_i^k}{r}\right)\right) \tag{3}$$

where P_i and P_i^k are XYZ positions of points, $\oplus$ stands for concatenation operation and r is the radius. Concatenating the directional position vectors with the relatively higher-dimensional features within a given neighborhood integrates them as features at the neighbor level. The incorporation of effective positional coding serves to augment the network's capacity to learn local features [37].

c) Global feature: The point locations P_i and point features f_i contain the overall shape and semantic information of the object, an MLP is employed to extract the global point features as defined in the following equation::

$$G_i = \mathrm{MLP}_g(P_i \oplus f_i). \tag{4}$$

A concatenation operation is executed on the extracted multifeatures to derive new features for each point within the neighborhood

$$F_i^k = \left(r_i^k \oplus G_i \oplus df_i^k \oplus f_i^k\right). \tag{5}$$

The final multifeature encoding layer yields a set of new neighboring point features, denoted as $F_i = \{F_i^1, \ldots, F_i^k, \ldots, F_i^K\}$, which effectively encodes the local structure of the central point P_i.

2) Multifeature Aggregation Layer: An efficient way of generating an effective overall feature embedding is through the fusion of extracted multifeatures. Point cloud-based Transformer [18], [19] has proven to be effective in aggregating features. Leveraging the full potential of multifeatures, a multilayer transformer block of Pointformer [18] is employed to aggregate neighborhood point information, as illustrated in the following equation:

$$F^{(l+1)} = \mathrm{Transformer}_l\left(F^{(l)}\right), \quad l = 0, 1, \ldots, L-1. \tag{6}$$

The structure of the model is shown in Fig. 2, employing the inner-product attention mechanism as described in the following equation:

$$f_i' = A\left(\sigma\left(\frac{f_i Q K^{\mathrm{T}} f_j^{\mathrm{T}}}{\sqrt{d}}\right) \times \mathrm{FFN}(f_i)\right) \quad \forall j \in P(p_i) \tag{7}$$

where d refers to the feature dimension of f. The multilayer transformer block is designed to take into account neighboring multifeatures and explore the complex local structure in greater depth. This process results in sampling point features that have stronger representation capabilities. To achieve this, a combination of max pooling operations and an MLP layer is used.

B. Semantic-Aware Downsampling Module

Existing downsampling strategies such as FPS and semantics-guided farthest point sampling (S-FPS) [12] are effective. Discriminative feature learning or helping point cloud downsampling based on the potential semantic information of each point is implemented using foreground segmentation strategies [23], [40], [41]. To oversee and guide this process, cross-entropy loss is applied as follows:

$$L_{\mathrm{cls}} = -\sum_{c=1}^{C}\left((1 - p_s)\log\left(1 - \hat{p}_s\right) + p_s \log\left(\hat{p}_s\right)\right). \tag{8}$$

Here, C represents the number of categories, p_s and $\hat{p}_s$ represent the label and predicted category probabilities, respectively. To perform the sampling process, the method selects the top k points with the highest semantic scores as the downsampled points. The corresponding points and features of these downsampled points are then used as the input for the next-stage feature encoding layer.

However, they fail to retain important foreground points during the downsampling process and do not consider density differences between object categories. As a result, small objects and distant point clouds are lost. To address this issue, we propose an adaptive semantic-aware downsampling module that preserves a greater number of valid foreground points. The module consists of the proposed cls_aware loss and cd_aware loss. To reduce memory and computational costs, the semantic-aware downsampling module employs a simplified network structure consisting of a two-layer MLP branch that outputs an importance score for each point. Although the semantic-aware downsampling module is more comprehensive, it does not theoretically incur more computational time in the inference phase, compared to the SASA [12] and IA-SSD [23].

In the context of point-based 3DOD, prioritizing the center of instances as pivotal for final object detection necessitates assigning higher scores to points proximal to the object's center [23], [26]. Therefore the distribution of points within the instance needs to be further considered so that more accurate keypoints can be selected. Besides, small objects have relatively fewer points and need to be considered for category variability. In response, we propose a novel cls_aware loss aimed at sampling more critical centroids, thereby imbuing the model with enhanced awareness of interpoint relationships

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within objects. This augmentation not only refines the object detection process but also ensures that the model's feature representation captures nuanced spatial dependencies within object structures.

As shown in (9), in the computation of the instance point mask score PS_i^j, higher importance scores are assigned to points that are closer to the center of the instance in the point cloud. The scores gradually decrease as points move away from the instance center, with background points receiving an importance score of 0. To ensure that foreground points of small objects in the scene receive higher importance scores relative to larger objects, such as cars, class-specific weights, denoted by the variable K_{cls}, are introduced for different categories

$$\mathrm{ps}_i^j = \exp\left[-\frac{K_{\mathrm{cls}}}{2}\left(\frac{\left(P_x^j - C_x^i\right)^2}{L^2+W^2} + \frac{\left(P_y^j - C_y^i\right)^2}{L^2+H^2} + \frac{\left(P_z^j - C_z^i\right)^2}{W^2+H^2}\right)\right]. \tag{9}$$

In the formula, $P^j(P_x^j, P_y^j, P_z^j)$ represents any point within instance i, $C_i(C_x^i, C_y^i, C_z^i)$ denotes the centroid coordinates of instance i, and L, W, and H represent the length, width, and height of instance i, respectively. Finally, the following equation introduces instance point scores as point masks into the $L_{\mathrm{cls_aware}}$ loss:

$$L_{\mathrm{cls_aware}} = -\sum_{c=1}^{C}\left(\mathrm{ps}^s \cdot p_s \log\left(\hat{p}_s\right) + (1-p_s)\log\left(1-\hat{p}_s\right)\right). \tag{10}$$

In the sampling stage, we use the point semantic information to focus on foreground objects. However, the current top-k approach selects an excessive number of points from easily recognizable objects which usually have higher foreground scores. This leads to the acquisition of sampling points that do not cover all objects, resulting in decreased detection performance. To address this challenge, downsampling all objects in the scene is necessary while minimizing the impact of noise. Inspiration can be drawn from the point cloud completion (PCN) task [42], [43]. Chamfer Distance (CD) [44] has been selected as the loss function to quantify the distribution disparity between the sampled output point cloud P_1 and the desired foreground point cloud P_2, as follows:

$$L_{\mathrm{cd_aware}} = \frac{1}{|P_1|}\sum_{x\in P_1}\min_{k\in P_2}\|x-k\| + \frac{1}{|P_2|}\sum_{k\in P_2}\min_{x\in P_1}\|k-x\| \tag{11}$$

where $||\cdot||$ represents the L1 norm, which is used to compute the distance between the two point cloud sets, and the whole CD is used to compute the average nearest point distance between the point cloud P_1 and P_2. Consideration of the distribution of points further ensures that the downsampled point contains as many instances of objects as possible.

C. Multitask Loss

The training of the proposed network model adopts an end-to-end strategy, wherein the multitask joint total loss comprises two downsampling losses, a classification loss, and a bounding box regression loss. This can be expressed as follows:

$$L_{\mathrm{total}} = \lambda_1 L_{\mathrm{cls_aware}} + \lambda_2 L_{\mathrm{cd_aware}} + \lambda_3 L_{\mathrm{cls}} + \lambda_4 L_{\mathrm{box}} \tag{12}$$

where λ is a hyperparameter whose values are all 1, and the multiple loss functions together constrain the training of the model.

D. Point Distribution Awareness-PV-RCNN

The point-voxel fusion method combines the efficient computation of a voxel-based backbone with accurate positional information extracted from raw point clouds. However, such methods, like PV-RCNN, use a limited set of keypoints obtained through FPS to summarize 3-D scenes. During keypoint selection, FPS tends to choose points at a greater distance to uniformly cover the entire point cloud. Our motivation is to retain more foreground points to capture valuable spatial and semantic information. Therefore, as illustrated in Fig. 3, the Point Distribution Awareness PV-RCNN (PDA-PV-RCNN) is further introduced, building upon the PV-RCNN framework. The module responsible for extracting keypoints has decided to forgo unbiased FPS sampling in favor of using PDANet. The use of PDANet helps obtain more valuable keypoints and their corresponding features. This makes it easier to combine features during the RoI grid pooling stage of PV-RCNN.

E. Point Distribution Awareness-CenterPoint

Some 3DOD methods consider objects as separate points on the BEV map and use these points to predict detection objects, such as CenterPoint. However, information of the point cloud can be lost during the voxel backbone and BEV representation process. Point-wise prediction serves as a complement to the BEV representation. Taking inspiration from this, we introduce a 3DOD framework on top of CenterPoint, termed Point Distribution Awareness-CenterPoint (PDA-CenterPoint). As shown in Fig. 4, this framework explores the Point-BEV fusion for CenterPoint. In the first stage, point cloud features are extracted using 3-D sparse convolution and a PDANet branch. The extracted point cloud features are then transformed into BEV features for fusion. In the second stage, the improved BEV features undergo 2-D feature convolution and detection head, resulting in the identification of detection boxes and their corresponding scores. It is worth noting that our main objective was to explore the complementary information between centroid point features in the PDANet and BEV features. Consequently, an attempt was made to concatenate the BEV features from different network branches.

Similar to PointPillar [45], [46], the last downsampled layer of the point cloud undergoes a pillarization process; in this process, the features within each grid are formed by fusing multiple point features within the corresponding region. Via the Point-to-BEV mapping operation described in the

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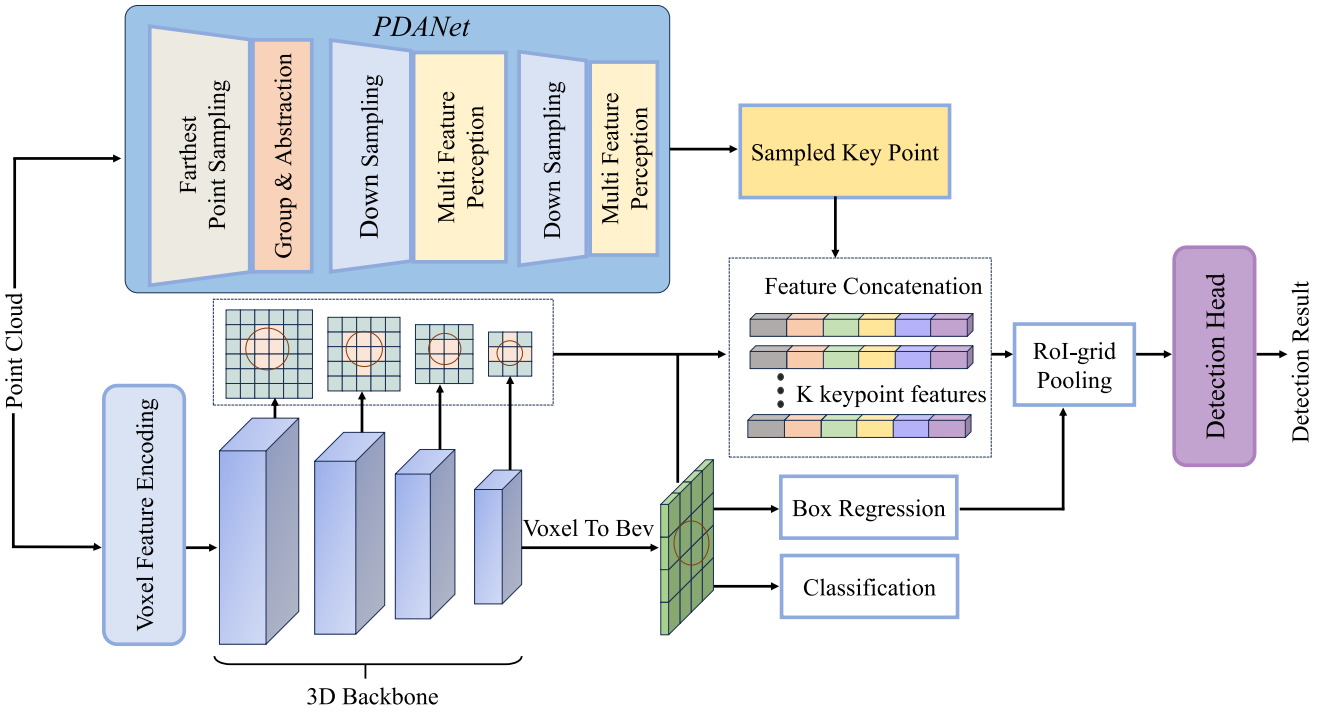


Fig. 3. Architecture of the PDA-PV-RCNN model that incorporates the PDANet.

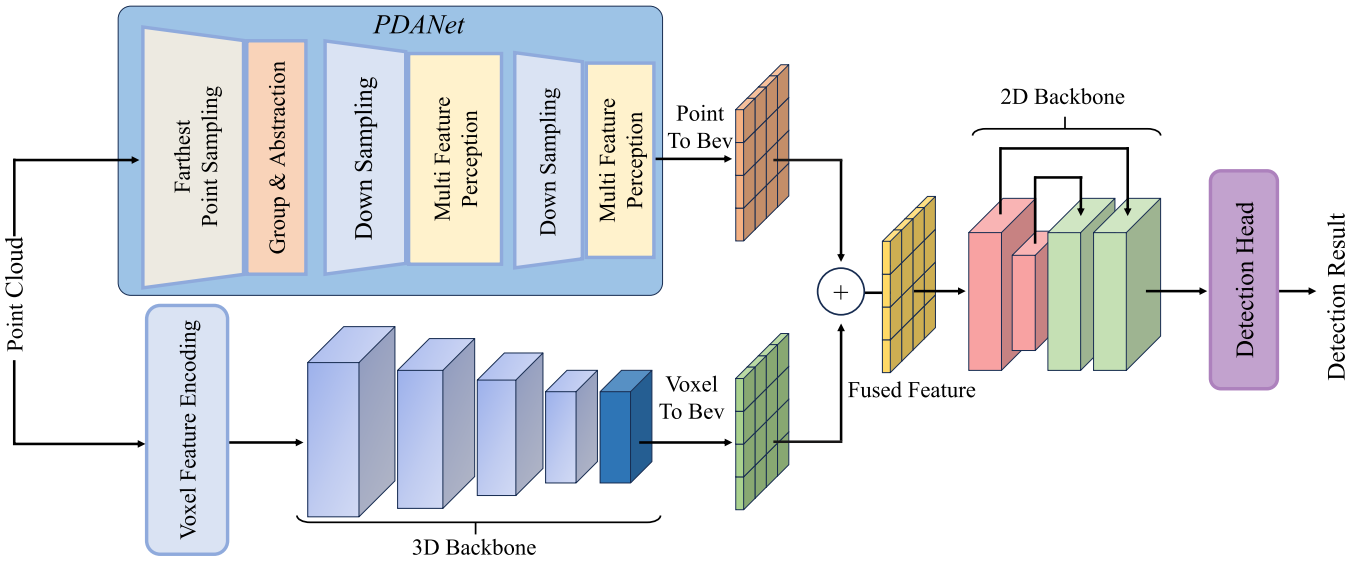


Fig. 4. Architecture of the PDA-CenterPoint model.

following equation, the new BEV feature F_{bev} is acquired:

$$F_{\text{bev}} = \frac{1}{N_{h,w}} \sum_{m=1}^{M} \mathbb{1}\left[s_m \in K_{h,w}\right] f_m^k \tag{13}$$

where h and w index the BEV feature maps $F_{h,w}$, $\mathbb{1}$ is an indicator function that determines which pillar each point belongs to, and M is the total number of points.

IV. Experiment and Results

A. Dataset and Implementation Details

All experiments were primarily conducted using the ONCE dataset for validation and analysis. Equipped with a 40-beam LiDAR and seven cameras, the ONCE dataset comprises 5000, 3000, and 8000 scenes annotated for training, validation, and test sets, respectively. Annotations cover diverse object categories, including vehicle (car, truck, bus), pedestrian, and cyclist. Average precision (AP) scores were obtained for the three categories of objects. The evaluation involved a comparison with other state-of-the-art algorithms on the ONCE dataset. Additionally, ablation experiments were conducted on the validation set. Assessments were conducted for each class at distinct distances: close-range, mid-range, and long-range, corresponding to 0–30 m, 30–50 m, and distances beyond 50 m. The mean average precision (mAP) was obtained by averaging the scores for vehicles, pedestrians, and cyclists.

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TABLE I
EVALUATION RESULTS OF OUR MODEL AND BASELINE MODELS ON THE ONCE TEST SET

Method	AP@50(%)			mAP(%)
	Vehicle	Pedestrian	Cyclist	
PointPillars [45]	69.52	17.28	49.63	45.47
SECOND [47]	69.71	26.09	59.92	51.90
PointRCNN [1]	52.00	8.73	34.02	31.58
CG-SSD [48]	68.00	52.81	67.50	62.77
IA-SSD [23]	71.38	36.86	64.80	57.68
PDA-SSD (Ours)	73.26	40.99	67.88	60.71
Improvement	*+1.88*	*+4.13*	*+3.08*	*+3.03*
PV-RCNN [24]	76.98	22.66	61.93	53.85
PDA-PV-RCNN (Ours)	75.19	33.76	66.39	58.45
Improvement	*-1.79*	*+11.10*	*+4.46*	*+4.60*
CenterPoint [25]	66.35	**51.80**	65.57	61.24
PDA-CenterPoint (Ours)	**78.82**	51.43	**71.92**	**67.39**
Improvement	*+12.47*	*-0.37*	*+6.35*	*+6.15*

Additionally, the KITTI dataset was used for further experiments. This dataset comprises a total of 7481 training samples and 7518 test samples, categorized into three main classes: cars, pedestrians, and cyclists. Each sample consists of data collected from a 64-beam LiDAR. The KITTI training samples are further divided into 3712 training samples and 3769 validation samples.

The OpenPCDet[1] open-source 3DOD library was employed for end-to-end training across multiple categories. The network underwent training for 80 epochs utilizing eight Nvidia RTX 3090 GPUs, with a combined batch size of 16. To ensure a fair comparison with other models, consistent configuration parameters were followed, including a weight decay of 0.01, aligned with the IA-SSD, PV-RCNN, and CenterPoint models.

B. Results and Analysis

1) Evaluation Using the Once Test Set: Results of the compared methods on the ONCE test dataset are presented in Table I. PDA-CenterPoint achieved the highest mAP of 67.39%. The PDANet demonstrates an improvement for all the three baseline models, achieving a maximum 11.10% increase in mAP from the original 3-D detector. Particularly, our PDANet exhibits greater advantages for pedestrian and cyclist detection. Compared to the baseline IA-SSD model, PDA-SSD achieves improvements of 1.88%, 4.13%, and 3.08% in mAP for the vehicle, pedestrian, and cyclist categories, respectively. PDA-SSD achieved a competitive AP of 67.88% in the cyclist category.

Similarly, the performances of PV-RCNN and CenterPoint are also improved with the integration of the PDANet. PDA-PV-RCNN achieved improvements of 11.10% and 4.46% in AP for the pedestrian and cyclist categories, respectively. Despite a decrease in AP for the vehicle category, it resulted in an overall mAP improvement of 4.60%. PDA-CenterPoint achieved improvements in the AP of all vehicles, pedestrians, and cyclists, ultimately reaching an optimal mAP of 67.39%. Overall, the results demonstrate that our model excels at capturing contextual information more effectively from LiDAR datasets.

2) Evaluation Using the Once Validation Set: For the assessment of our method's performance at different distances, additional evaluations were conducted using the ONCE validation set. As illustrated in Table II, PDANet demonstrated detection improvements across all baseline models, with PDA-CenterPoint achieving the highest mAP of 66.72%.

In the car category, PDA-SSD outperforms the baseline IA-SSD method in AP by 1.85%, 4.23%, and 6.12% at different distances. For the pedestrian and cyclist categories, PDA-SSD consistently improves IA-SSD [23] by at least 3.81% AP at various distances. As the object-to-sensor distance increases, our approach exhibits greater precision compared to IA-SSD. In Fig. 5, the detection results of IA-SSD and PDA-SSD models are presented, showing that false positives have been reduced and more true positives are detected.

The performance of PDA-PV-RCNN exhibits a slight decrease in detecting vehicles, but it brings about an improvement in AP as much as 13.47% for pedestrians. Improvements of 9.21% and 5.07% in overall AP were achieved for pedestrian and cyclists, respectively.

PDA-CenterPoint enhances the overall AP of CenterPoint by 1.89%, 1.88%, and 0.41% for the three categories, respectively. As the distance increases, our model continues to outperform the corresponding baseline models. This is primarily attributed to the semantic-aware downsampling module aims to preserve the overall geometric information of objects, minimizing the generation of outliers. Additionally, the MFAP module further extracts geometric and semantic information from objects. For instance, the density-aware perception module adapts to changes in local density at different distances. Ultimately, our model learns more accurate 3-D information as a whole.

3) Evaluation Using the KITTI Validation Set: The performances of PDA-SSD and PDA-PV-RCNN were assessed on the KITTI validation set, with results reported on the moderately difficult evaluation metric. Similar to CenterPoint, a PDA-CasA-V was formed based on PDANet and CasA-V [54] as a voxel-based model and tested on the same data. As shown in Table III, the detection results of all three models for both pedestrian and cyclist categories were improved. PDA-SSD achieved an AP of 61.09% and 72.84%, representing enhancements of 7.41% and 3.49% for pedestrians and cyclists, respectively. In addition, the AP of car was also slightly improved. PDA-PV-RCNN demonstrated respective enhancements of 6.27% and 1.13% of AP in detecting pedestrians and cyclists, as shown in Fig. 7, further substantiating the feature enhancement effect of the PDANet on RoI grid pooling. Although the detection of cars dropped by 2.09%, the overall performance was improved. PDA-CasA-V increased the pedestrian detection AP by 1.35% and that of cyclists by a marginal 0.22%, resulting in an enhanced overall performance, even the car AP was also decreased marginally

[1]https://github.com/open-mmlab/OpenPCDet

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TABLE II
EVALUATION RESULTS OF OUR MODEL AND BASELINE MODELS ON THE ONCE VALIDATION SET

Method	Vehicle (AP@50)(%)				Pedestrian (AP@50)(%)				Cyclist (AP@50)(%)				mAP(%)
	Overall	0–30 m	30–50 m	50 m-inf	Overall	0–30 m	30–50 m	50 m-inf	Overall	0–30 m	30–50 m	50 m-inf	
PointRCNN [1]	52.09	74.45	40.89	16.81	4.28	6.17	2.40	0.91	29.84	46.03	20.94	5.46	28.74
PointPillars [45]	68.57	80.86	62.07	47.04	17.63	19.74	15.15	10.23	46.81	58.33	40.32	25.86	44.34
SECOND [47]	71.19	84.04	63.02	47.25	26.44	29.33	24.05	18.05	58.04	69.96	52.43	34.61	51.89
CG-SSD [48]	67.60	80.22	61.23	44.77	51.50	**58.72**	**43.36**	**27.76**	65.79	76.27	60.84	43.35	61.63
GD-MAE [49]	76.79	88.01	71.70	55.60	48.84	58.70	37.30	25.72	69.14	80.29	64.58	45.14	64.92
PARTNER [50]	68.07	N/A	N/A	N/A	**53.45**	N/A	N/A	N/A	65.93	N/A	N/A	N/A	63.15
IA-SSD [23]	70.30	83.01	62.84	47.01	39.82	47.45	32.75	18.99	62.17	73.78	56.31	39.53	57.43
PDA-SSD	74.10	84.86	67.07	53.13	44.67	51.26	38.55	23.44	66.24	77.71	60.26	44.22	61.67
Improvement	*+3.80*	*+1.85*	*+4.23*	*+6.12*	*+4.85*	*+3.81*	*+5.80*	*+4.45*	*+4.07*	*+3.93*	*+3.95*	*+4.69*	*+4.24*
PV-RCNN [24]	77.77	**89.39**	72.55	58.64	23.50	25.61	22.84	17.27	59.37	71.66	52.58	36.17	53.55
PDA-PV-RCNN	76.79	87.57	70.81	55.80	32.71	39.08	28.84	18.05	64.44	75.43	58.57	39.97	57.98
Improvement	*-0.98*	*-1.82*	*-1.74*	*-2.84*	*+9.21*	*+13.47*	*+6.00*	*+0.78*	*+5.07*	*+3.77*	*+5.99*	*+3.80*	*+4.43*
CenterPoint [25]	66.79	80.10	59.55	43.39	49.90	56.24	42.61	26.27	63.45	74.28	57.94	41.48	60.05
CenterPoint* [25]	78.08	86.43	73.85	62.41	47.91	55.84	38.69	24.16	69.98	**80.67**	64.71	50.29	65.32
PDA-CenterPoint	**79.97**	88.73	**74.72**	**62.63**	49.79	57.37	41.52	26.18	**70.39**	80.05	**65.35**	**51.46**	**66.72**
Improvement	*+1.89*	*+2.30*	*+0.87*	*+0.22*	*+1.88*	*+1.53*	*+2.83*	*+2.02*	*+0.41*	*-0.62*	*+1.18*	*+1.17*	*+1.40*

Note: * represents the reproduction of CenterPoint.

TABLE III
EVALUATION RESULTS OF OUR MODEL AND BASELINE MODELS ON THE KITTI VALIDATION SET

	Method	Type	Car (IoU=0.7)	Pedestrain (IoU=0.5)	Cyclist (IoU=0.5)
Voxel-based	VoxelNet [51]	1-stage	65.11	33.69	48.36
	SECOND [47]	1-stage	78.62	52.99	67.13
	PointPillars [45]	1-stage	77.28	52.30	62.68
	Voxel-RCNN [52]	2-stage	65.50	53.40	47.70
	MF-Net [53]	1-stage	77.09	58.19	55.61
	CasA-V [54]	2-stage	**86.58**	N/A	N/A
	CasA-V* [54]	2-stage	86.53	66.28	74.00
	PDA-CasA-V	2-stage	86.37	**67.63**	**74.22**
	Improvement		*-0.16*	*+1.35*	*+0.22*
Point-Voxel	PP-RCNN [55]	2-stage	78.63	N/A	67.68
	VIC-Net [56]	1-stage	79.25	N/A	N/A
	PV-RCNN [24]	2-stage	81.13	54.47	70.53
	PDA-PV-RCNN	2-stage	79.04	60.74	71.66
	Improvement		*-2.09*	*+6.27*	*+1.13*
Point-based	PointRCNN [1]	2-stage	78.57	54.08	71.41
	3DSSD [22]	1-stage	78.45	54.31	70.48
	CA-SSD [57]	1-stage	80.76	55.24	72.70
	IA-SSD* [23]	1-stage	79.20	53.68	69.35
	PDA-SSD	1-stage	79.27	61.09	72.84
	Improvement		*+0.07*	*+7.41*	*+3.49*

Note: * represents the reproduction of model.

by 0.16%. The results of the three models further demonstrate the plug-and-play nature of our method and its effectiveness on small objects. Even for moderate difficulty, our PDANet ensured an improved perception of pedestrians and cyclists despite a small number of points.

C. Ablation Study

1) Effect of Different Modules: For evaluating the impact of different modules, a progressive addition of our modules to the baseline network was performed, as presented in Table IV. The MFAP module resulted in improvements of 3.69%, 3.05%, and 2.51% of AP for vehicles, pedestrians, and cyclists, respectively. The semantic-aware downsampling module consists of the proposed cls_aware loss and cd_aware loss. The loss of cls_aware improves the cyclist's AP by 1.37% and the mAP by 0.75%. However, pedestrian AP in the 50 m-inf range decreased by 0.62%, possibly due to the limited number of distant pedestrian point clouds, making the perception challenging. The cd_aware loss mitigated this issue to some extent, contributing to an increase of 0.4% of AP for pedestrians at 50 m-inf and improving overall AP performance for both pedestrians and cyclists. Overall, the semantic-aware downsampling module further improves the AP of each category, ultimately improving the mAP by 1.16%. The MFAP module and the semantic-aware downsampling module complement each other, resulting in an mAP of 61.67%.

For the ONCE dataset, the input point clouds in IA-SSD and PDA-SSD are fed into four different SA layers and sampling

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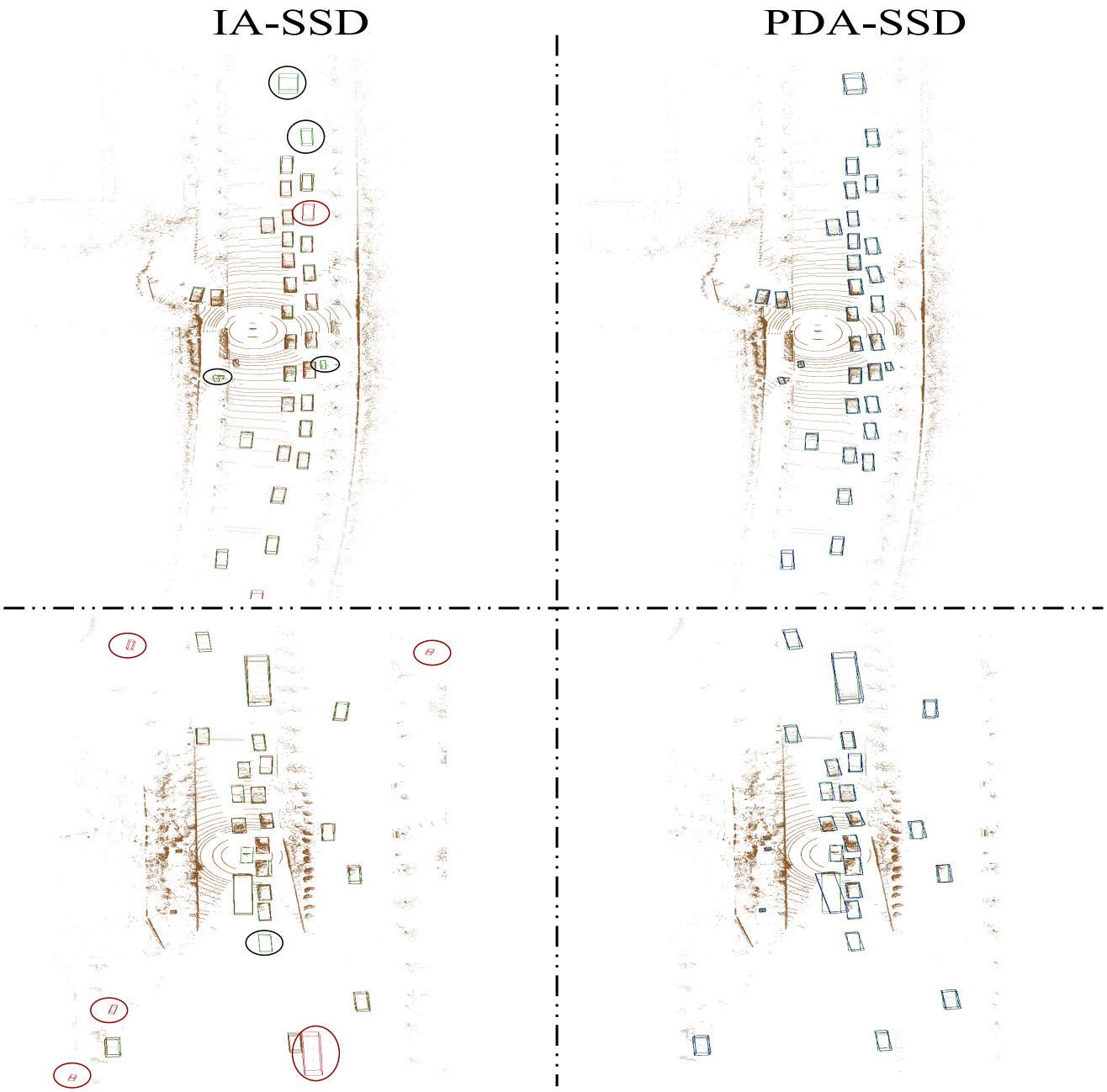


Fig. 5. Visualization of (left) IA-SSD and (right) PDA-SSD model detection results on ONCE dataset. The green boxes represent the ground truth, while the red ones show the predicted bounding boxes. Falsely detected objects are marked with red ellipses and the objects detected only by PDA-SSD are highlighted with black ovals.

layers for feature extraction, and each downsampling layer yields 16 384, 4096, 2048, and 1024 points of information, including point coordinates and other extracted features. Given the constrained semantic depth inherent in the initial layers, the D-FPS strategy is used in the first two sampling layers and the proposed semantic-aware downsampling module is used in the other layers. Fig. 6 demonstrates the results of sampling scores at different layers of IA-SSD and PDA-SSD, followed by the selection of 1024 representative points for input into the subsequent prediction module. In this scenario, IA-SSD exhibits inefficiency in discerning background objects, such as railings, amidst multiple downsampling stages, consequently resulting in false detections. Moreover, for remote sparse vehicles, IA-SSD experiences a progressive degradation in the fidelity of the vehicle's point cloud representation during multilayer downsampling. Conversely, PDA-SSD adeptly preserves both the geometric and semantic attributes of the object's point cloud, thereby capturing the salient features essential for accurate object detection.

2) Effect of Different Features in the MFAP Module: The impact of different feature modules are presented in Table V, where various features were successively incorporated into

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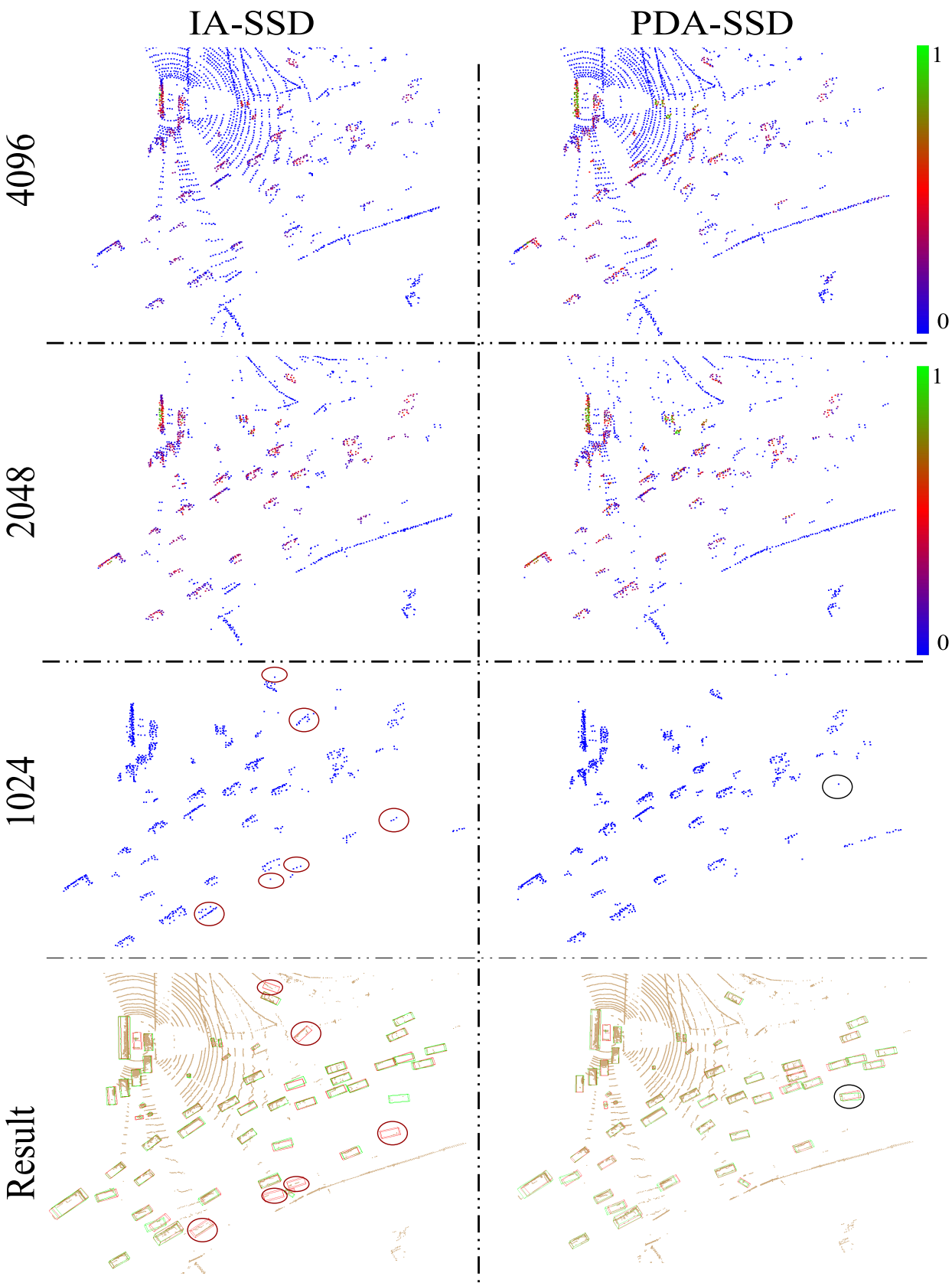


Fig. 6. Visualization of (left) IA-SSD and (right) PDA-SSD model detection results on ONCE dataset. Three sampling layers in PDA-Net give 4096, 2048, and 1024 points after sampling. The green boxes represent the ground truth, while the red ones show the predicted bounding boxes. Falsely detected objects are marked with red ellipses, and the objects detected by PDA-SSD are highlighted with black ovals.

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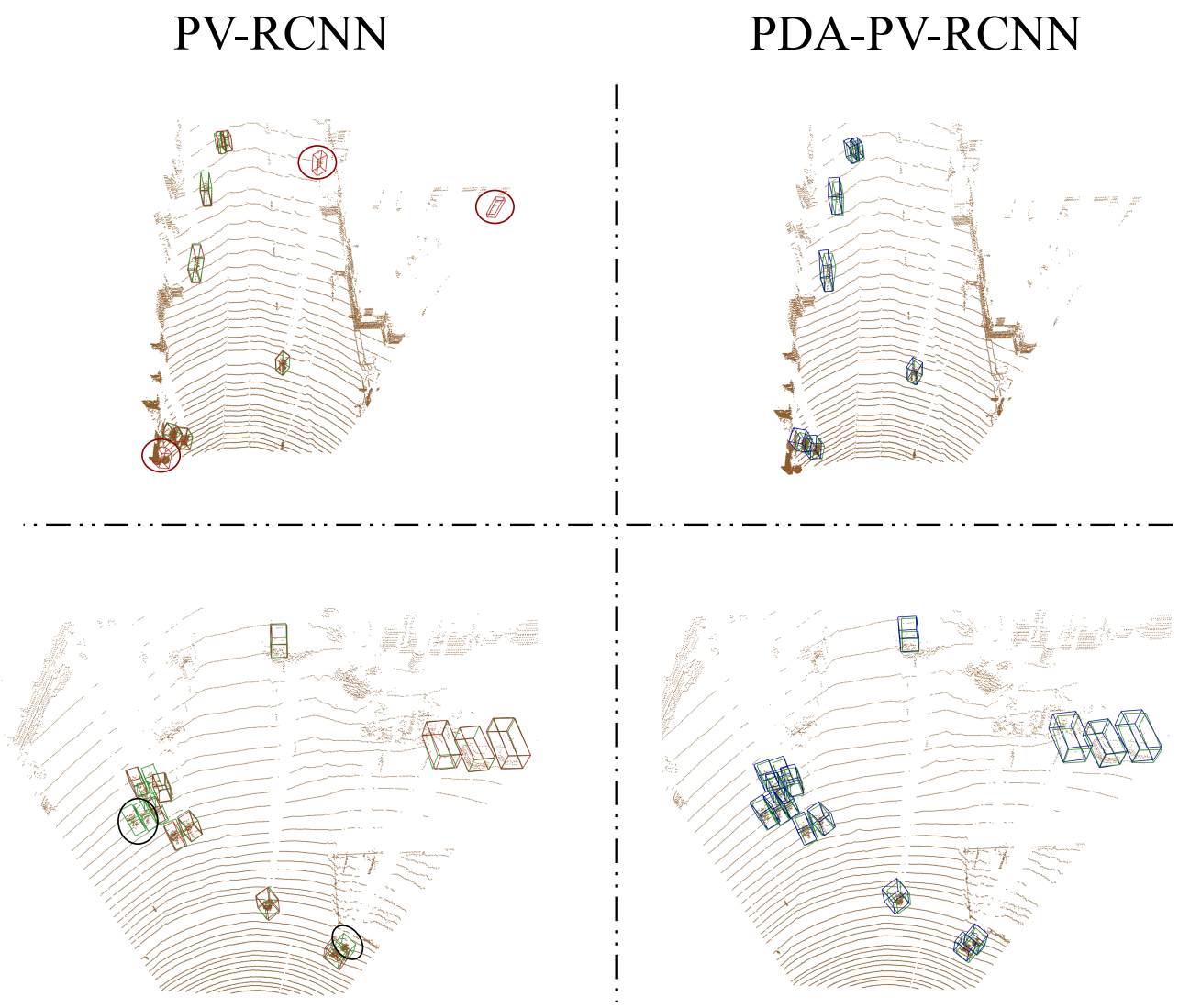


Fig. 7. Visualization of (left) PV-RCNN and (right) PDA-PV-RCNN model detection results on KITTI dataset. The green boxes represent the ground truth, while the red ones show the predicted bounding boxes. Falsely detected objects are marked with red ellipses, and the objects detected only by PDA-PV-RCNN are highlighted with black ovals.

TABLE IV
EFFECT OF DIFFERENT MODULES IN PDA-SSD ON THE ONCE VALIDATION SET

Method	Vehicle (AP@50)(%)				Pedestrian (AP@50)(%)				Cyclist (AP@50)(%)				mAP(%)
	Overall	0–30 m	30–50 m	50 m-inf	Overall	0–30 m	30–50 m	50 m-inf	Overall	0–30 m	30–50 m	50 m-inf	
Baseline	70.30	83.01	62.84	47.01	39.82	47.45	32.75	18.99	62.17	73.78	56.31	39.53	57.43
+MFAP	73.99	84.77	66.94	52.37	42.87	49.17	36.99	23.66	64.68	76.10	59.87	41.49	60.51
+L_{cls_aware}	**74.57**	**84.91**	**68.81**	**54.83**	43.16	49.53	37.39	23.04	66.05	76.40	**60.70**	42.99	61.26
+L_{cd_aware}	74.10	84.86	67.07	53.13	**44.67**	**51.26**	**38.55**	**23.44**	**66.24**	**77.71**	60.26	**44.22**	**61.67**

the multifeature encoding layer. It is evident that our local spatial feature enhances the detection performance for objects at distances of 30–50 m and 50 m-inf, indicating that effective positional and directional information at longer distances is beneficial for object perception. Considering the limited number of point clouds for pedestrians and their sparse and irregular geometric structure, our proposed Gaussian density feature captures fine-grained variations in local point cloud density, resulting in a 4.40% improvement of AP in the 30–50 m range. This demonstrates that our density perception can handle objects with multiple postures. Ultimately, the MFAP module enhances the mAP to 60.51%. Specifically, it improves the AP of cars, pedestrians, and cyclists by 3.69%, 3.05%, and 2.51%, respectively. The AP at different distances was

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TABLE V
EFFECT OF DIFFERENT FEATURES IN THE MFAP MODULE ON THE ONCE VALIDATION SET

Method	Vehicle (AP@50)(%)				Pedestrian (AP@50)(%)				Cyclist (AP@50)(%)				mAP(%)
	Overall	0–30 m	30–50 m	50 m-inf	Overall	0–30 m	30–50 m	50 m-inf	Overall	0–30 m	30–50 m	50 m-inf	
Baseline	70.30	83.01	62.84	47.01	39.82	47.45	32.75	18.99	62.17	73.78	56.31	39.53	57.43
+LSF	72.48	84.62	66.75	51.31	39.75	45.34	33.99	19.85	63.02	73.50	57.99	40.20	58.42
+GDF	72.42	84.69	64.99	50.58	43.71	49.80	**38.39**	**23.83**	63.55	74.18	57.77	40.32	59.89
+GF	**73.99**	**84.77**	**66.94**	**52.37**	**42.87**	**49.17**	36.99	23.66	**64.68**	**76.10**	**59.87**	**41.49**	**60.51**

Note: LSF: Local Spatial Feature, GDF: Gaussian Density feature, GF: Global Feature.

TABLE VI
EFFECT OF THE PDANET ON DIFFERENT MODELS USING THE ONCE VALIDATION SET

Method	Vehicle (AP@50)(%)				Pedestrian (AP@50)(%)				Cyclist (AP@50)(%)				mAP(%)
	Overall	0–30 m	30–50 m	50 m-inf	Overall	0–30 m	30–50 m	50 m-inf	Overall	0–30 m	30–50 m	50 m-inf	
PV-RCNN	**77.77**	**89.39**	**72.55**	**58.64**	23.50	25.61	22.84	17.27	59.37	71.66	52.58	36.17	53.55
PV-RCNN*	74.98	86.71	69.18	56.56	32.49	38.96	27.92	18.61	62.04	74.42	55.30	37.27	56.51
PDA-PV-RCNN	76.79	87.57	70.81	55.80	**32.71**	**39.08**	**28.84**	**18.05**	**64.44**	**75.43**	**58.57**	**39.97**	**57.98**
CenterPoint	78.08	86.43	73.85	62.41	47.91	55.84	38.69	24.16	69.98	**80.67**	64.71	50.29	65.32
CenterPoint*	78.14	88.10	74.15	62.22	48.68	55.61	40.74	**27.22**	69.99	79.75	**65.44**	49.06	65.60
PDA-CenterPoint	**79.97**	**88.73**	**74.72**	**62.63**	**49.79**	**57.37**	**41.52**	26.18	**70.39**	80.05	65.35	**51.46**	**66.72**

Note: * represents the integration of the original downsampling backbone network of IA-SSD.

TABLE VII
EFFECT OF MFAP MODULE ON DIFFERENT MODELS USING THE ONCE VALIDATION SET

Method	Vehicle (AP@50)(%)				Pedestrian (AP@50)(%)				Cyclist (AP@50)(%)				mAP(%)
	Overall	0–30 m	30–50 m	50 m-inf	Overall	0–30 m	30–50 m	50 m-inf	Overall	0–30 m	30–50 m	50 m-inf	
PointRCNN	52.09	74.45	40.89	16.81	4.28	6.17	2.40	0.91	29.84	46.03	20.94	5.46	28.74
+ MFAP	**52.19**	**74.47**	**41.30**	**18.66**	**5.91**	**9.15**	**3.37**	**1.45**	**33.83**	**51.89**	**24.82**	**8.47**	**30.65**
IA-SSD	70.30	83.01	62.84	47.01	39.82	47.45	32.75	18.99	62.17	73.78	56.31	39.53	57.43
+ MFAP	**73.99**	**84.77**	**66.94**	**52.37**	**42.87**	**49.17**	**36.99**	**23.66**	**64.68**	**76.10**	**59.87**	**41.49**	**60.51**

increased range from 1.72% to 5.36%, demonstrating that our MFAP module can better extract local shape features from point clouds while preserving shape invariance.

3) Effect of PDANet Auxiliary Module: By leveraging the PDANet point downsampling feature perception module, existing point-voxel-based models (PV-RCNN) and voxel-based models (CenterPoint) are enhanced. As shown in Table VI, we first integrated the downsampling backbone network from IA-SSD as a branch into PV-RCNN and CenterPoint, resulted in an increase of 2.96% and 0.3% mAP for PV-RCNN and CenterPoint, respectively, validating the effective information complement provided by the point-wise feature. Similarly, the PDANet was integrated as a branch into PV-RCNN and CenterPoint, demonstrating a more heightened improvement in the existing networks. Despite a decrease in AP for the vehicle category, PDA-PV-RCNN demonstrates further improvements in the pedestrian and cyclist categories, resulting in 57.98% of mAP. Compared to the CenterPoint detector, PDA-CenterPoint has an increase in AP across various distances, contributing to a 1.4% improvement in mAP. This showcases that the PDA module effectively acquires critical points and features, thereby enhancing the detection capability of existing detectors.

4) Generality of the MFAP Module: The MFAP module comprises a series of point convolution modules that can be easily integrated as a plug-and-play component to replace the network's PointNet++ feature aggregation layer in any point-based 3DOD model. To validate its generality, we selected IA-SSD and PointRCNN for experimentation on the ONCE validation set. As shown in Table VII, using the MFAP module resulted in improvements of 1.91% and 3.08% of mAP for PointRCNN and IA-SSD, respectively. Combining our proposed multifeature aggregation module, the detection accuracy of objects at various distances showed a certain level of enhancement, confirming the module's robustness in perceiving local point cloud variations.

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V. Discussion

The experiments demonstrate that our PDANet effectively improves the overall performance of point-based (IA-SSD), voxel-based (PV-RCNN), and point-voxel-based (CenterPoint) models. There is a notable improvement in the detection of pedestrians and cyclists. In contrast, the enhancement for vehicle detection in our method is relatively modest, primarily because the existing detectors generally exhibit low false positives when dealing with relatively easily detectable objects. Overall, our approach positively influences the detection of small objects.

Despite the improvements in all three types of models, the overall accuracy of 3DOD utilizing single-frame point clouds from the ONCE and KITTI datasets is still low. There are three main reasons. 1) Objects present a localization challenge owing to the loss of shape information resulting from external occlusion, self-occlusion, and the sparsity of distant points in the LiDAR point cloud. 2) The detection of the object's orientation brings certain challenges. Most objects like cars and cyclists are typically axis-aligned, with minimal changes in orientation. Meanwhile, the orientation of a pedestrian is challenging to determine. Whereas, existing models focus primarily on object shape and position rather than object orientation. 3) In a single-frame point cloud, there occurs a geometric similarity between background and foreground objects in complex road environments, leading to the detection of false positives.

We believe two aspects can be further investigated to address the above issues. On the one hand, complementing LiDAR with cameras can bring extra information, which circumvents the risk of over-reliance on a single sensor. Multimodal feature fusion of LiDAR points and images has been proven to improve object detection. On the other hand, multiframe point clouds describe objects over time from different perspectives. Spatiotemporal point cloud features extracted from multiframe point clouds can capture more comprehensive object information.

Moreover, Transformer architecture has been extensively applied in 3DOD frameworks that are point [29], voxel [58], or point-voxel fusion [59] based. Pointformer [18] proposes a local transformer to learn features within a local region and a global transformer to learn global scene context features. Conversely, GAPSA [19] introduces gateway attention to enhance local semantic features. In the context of this study, the employed Transformer module primarily aims at bolstering fine-grained perception capabilities. Considering the balance between speed and accuracy, a Transformer module with faster computation is the goal of our future work.

VI. Conclusion

In this article, we introduce a 3DOD network named PDANet, comprising two modules, a MFAP module and a semantic-aware downsampling module. The MFAP module is designed to acquire richer contextual information, leading to improved accuracy of objects at varying distances. By incorporating the semantic-aware downsampling module, the identification of critical foreground points becomes more effective. We integrate the PDANet as a branch into existing point-based (IA-SSD), point-voxel-based (PV-RCNN), and voxel-based (Centerpoint) detectors and validate the impact of our modules on the performance using the ONCE and KITTI datasets. The results demonstrate the effectiveness of the proposed PDANet, especially for the detection of small objects.

References

[1] S. Shi, X. Wang, and H. Li, "PointRCNN: 3D object proposal generation and detection from point cloud," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 770–779.

[2] Y. Guo, H. Wang, Q. Hu, H. Liu, L. Liu, and M. Bennamoun, "Deep learning for 3D point clouds: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 12, pp. 4338–4364, Dec. 2021.

[3] R. Qian, X. Lai, and X. Li, "3D object detection for autonomous driving: A survey," *Pattern Recognit.*, vol. 130, Oct. 2022, Art. no. 108796.

[4] J. Mao, S. Shi, X. Wang, and H. Li, "3D object detection for autonomous driving: A comprehensive survey," *Int. J. Comput. Vis.*, vol. 131, no. 8, pp. 1909–1963, Aug. 2023.

[5] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? The KITTI vision benchmark suite," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 3354–3361.

[6] J. Mao et al., "One million scenes for autonomous driving: ONCE dataset," in *Proc. Adv. Neural Inf. Process. Syst.*, Jun. 2021, pp. 1–21.

[7] P. Sun et al., "Sparse R-CNN: End-to-end object detection with learnable proposals," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14449–14458.

[8] Y. Wan, Z. Liao, J. Liu, W. Song, H. Ji, and Z. Gao, "Small object detection leveraging density-aware scale adaptation," *Photogramm. Rec.*, vol. 38, no. 182, pp. 160–175, Jun. 2023.

[9] Y. Li, R. Bu, M. Sun, W. Wu, X. Di, and B. Chen, "PointCNN: Convolution on χ-transformed points," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2018, pp. 820–830.

[10] R. Q. Charles, H. Su, M. Kaichun, and L. J. Guibas, "PointNet: Deep learning on point sets for 3D classification and segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 77–85.

[11] C. R. Qi, L. Yi, H. Su, and L. J. Guibas, "PointNet++: Deep hierarchical feature learning on point sets in a metric space," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, Dec. 2017, pp. 5100–5109.

[12] C. Chen, Z. Chen, J. Zhang, and D. Tao, "SASA: Semantics-augmented set abstraction for point-based 3D object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 1, pp. 221–229.

[13] M. Chen et al., "A semantic segmentation method for vehicle-borne laser scanning point clouds in motorway scenes," *Photogramm. Rec.*, vol. 38, no. 182, pp. 94–117, Jun. 2023.

[14] C. Zhang and H. Fan, "An improved multi-task pointwise network for segmentation of building roofs in airborne laser scanning point clouds," *Photogramm. Rec.*, vol. 37, no. 179, pp. 260–284, Sep. 2022.

[15] Z. Zhang et al., "DMSC-Net: A deep multi-scale context network for 3D object detection of indoor point clouds," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 122, Aug. 2023, Art. no. 103454.

[16] H. Yu, J. Su, G. Cai, Y. Piao, N. Liu, and M. Huang, "3DSAC: Size adaptive clustering for 3D object detection in point clouds," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 118, Apr. 2023, Art. no. 103231.

[17] Z. Huang, Y. Wang, J. Wen, P. Wang, and X. Cai, "An object detection algorithm combining semantic and geometric information of the 3D point cloud," *Adv. Eng. Informat.*, vol. 56, Apr. 2023, Art. no. 101971.

[18] X. Pan, Z. Xia, S. Song, L. E. Li, and G. Huang, "3D object detection with pointformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7459–7468.

[19] H. Liu, J. Du, Y. Zhang, and H. Zhang, "Extracting geometric and semantic point cloud features with gateway attention for accurate 3D object detection," *Eng. Appl. Artif. Intell.*, vol. 123, Aug. 2023, Art. no. 106227.

[20] Z. Liu, Z. Zhang, Y. Cao, H. Hu, and X. Tong, "Group-free 3D object detection via transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 2929–2938.

[21] S. Huang, G. Cai, Z. Wang, Q. Xia, and R. Wang, "SSA3D: Semantic segmentation assisted one-stage three-dimensional vehicle object detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 9, pp. 14764–14778, Sep. 2022.

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[22] Z. Yang, Y. Sun, S. Liu, and J. Jia, "3DSSD: Point-based 3D single stage object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11037–11045.

[23] Y. Zhang, Q. Hu, G. Xu, Y. Ma, J. Wan, and Y. Guo, "Not all points are equal: Learning highly efficient point-based detectors for 3D LiDAR point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 18931–18940.

[24] S. Shi et al., "PV-RCNN: Point-voxel feature set abstraction for 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10526–10535.

[25] T. Yin, X. Zhou, and P. Krähenbühl, "Center-based 3D object detection and tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 11779–11788.

[26] C. R. Qi, O. Litany, K. He, and L. Guibas, "Deep Hough voting for 3D object detection in point clouds," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9276–9285.

[27] J. Yang et al., "DBQ-SSD: Dynamic ball query for efficient 3D object detection," 2022, *arXiv:2207.10909*.

[28] T. Xie et al., "FARP-Net: Local-global feature aggregation and relation-aware proposals for 3D object detection," *IEEE Trans. Multimedia*, vol. 26, no. 8, pp. 1027–1040, 2023.

[29] H. Shenga et al., "Improving 3D object detection with channel-wise transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 2723–2732.

[30] E. R. Corral-Soto, A. Grandhi, Y. Y. He, M. Rochan, and B. Liu, "Improving LiDAR 3D object detection via range-based point cloud density optimization," 2023, *arXiv:2306.05663*.

[31] J. S. K. Hu, T. Kuai, and S. L. Waslander, "Point density-aware voxels for LiDAR 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8459–8468.

[32] Q. Xia et al., "3-D HANet: A flexible 3-D heatmap auxiliary network for object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5701113.

[33] S. Wang, K. Lu, J. Xue, and Y. Zhao, "DA-Net: Density-aware 3D object detection network for point clouds," *IEEE Trans. Multimedia*, vol. 14, no. 8, pp. 1–14, 2023.

[34] D. Schinagl, G. Krispel, H. Possegger, P. M. Roth, and H. Bischof, "OccAM's laser: Occlusion-based attribution maps for 3D object detectors on LiDAR data," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1131–1140.

[35] J. Yang et al., "Modeling point clouds with self-attention and Gumbel subset sampling," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3318–3327.

[36] M. Wang, Q. Chen, and Z. Fu, "LSNet: Learned sampling network for 3D object detection from point clouds," *Remote Sens.*, vol. 14, no. 7, p. 1539, Mar. 2022.

[37] Q. Hu et al., "RandLA-Net: Efficient semantic segmentation of large-scale point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11105–11114.

[38] E. Parzen, "On estimation of a probability density function and mode," *Ann. Math. Statist.*, vol. 33, no. 3, pp. 1065–1076, Sep. 1962.

[39] W. Wu, Z. Qi, and L. Fuxin, "PointConv: Deep convolutional networks on 3D point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9613–9622.

[40] C. He, H. Zeng, J. Huang, X.-S. Hua, and L. Zhang, "Structure aware single-stage 3D object detection from point cloud," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11870–11879.

[41] T. D. Ngo, B.-S. Hua, and K. Nguyen, "ISBNet: A 3D point cloud instance segmentation network with instance-aware sampling and box-aware dynamic convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 13550–13559.

[42] W. Yuan, T. Khot, D. Held, C. Mertz, and M. Hebert, "PCN: Point completion network," in *Proc. Int. Conf. 3D Vis. (3DV)*, Sep. 2018, pp. 728–737.

[43] P. Xiang et al., "SnowflakeNet: Point cloud completion by snowflake point deconvolution with skip-transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 5479–5489.

[44] H. Fan, H. Su, and L. Guibas, "A point set generation network for 3D object reconstruction from a single image," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2463–2471.

[45] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, and O. Beijbom, "PointPillars: Fast encoders for object detection from point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 12689–12697.

[46] Z. Luo et al., "Exploring point-BEV fusion for 3D point cloud object tracking with transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 9, pp. 5921–5935, Sep. 2024.

[47] Y. Yan, Y. Mao, and B. Li, "SECOND: Sparsely embedded convolutional detection," *Sensors*, vol. 18, no. 10, p. 3337, Oct. 2018.

[48] R. Ma et al., "CG-SSD: Corner guided single stage 3D object detection from LiDAR point cloud," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 33–48, Sep. 2022.

[49] H. Yang et al., "GD-MAE: Generative decoder for MAE pre-training on LiDAR point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 9403–9414.

[50] M. Nie et al., "PARTNER: Level up the polar representation for LiDAR 3D object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 3801–3813.

[51] Y. Zhou and O. Tuzel, "VoxelNet: End-to-end learning for point cloud based 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4490–4499.

[52] J. Deng, S. Shi, P. Li, W. Zhou, Y. Zhang, and H. Li, "Voxel R-CNN : Towards high performance voxel-based 3D object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 1201–1209.

[53] Z. Meng, G. Luo, Q. Yuan, J. Li, and F. Yang, "MF-Net: Meta fusion network for 3D object detection," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2021, pp. 1–8.

[54] H. Wu, J. Deng, C. Wen, X. Li, C. Wang, and J. Li, "CasA: A cascade attention network for 3-D object detection from LiDAR point clouds," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5704511.

[55] J. Tu, P. Wang, and F. Liu, "PP-RCNN: Point-pillars feature set abstraction for 3D real-time object detection," in *Proc. Int. Joint Conf. Neural Netw.*, Jul. 2021, pp. 1–8.

[56] T. Jiang, N. Song, H. Liu, R. Yin, Y. Gong, and J. Yao, "VIC-Net: Voxelization information compensation network for point cloud 3D object detection," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2021, pp. 13408–13414.

[57] J. Liu, H. Cheng, G. Xiao, and X. Han, "CA-SSD: Channel-independent rare class attention for 3D object detection," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jun. 2023, pp. 1–8.

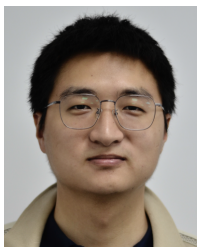
[58] L. Fan et al., "Embracing single stride 3D object detector with sparse transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8448–8458.

[59] H. Yang et al., "PVT-SSD: Single-stage 3D object detector with point-voxel transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 13476–13487.



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